UNIVERSITY OF GREENWICH
FACULTY OF ENGINEERING AND SCIENCE
SCHOOL OF COMPUTING AND MATHEMATICAL SCIENCES

COMP-1817 Information Visualisation and Big Data Coursework

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DECLARATION OF AI USE

Please complete this part when you have used AI during the process of undertaking this assignment to acknowledge the ways in which you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – YES
- To create an outline of the topic – NO
- To explain concepts – YES
- To support my use of language – YES

1. Task A: Big Data Project Analysis

Task A.1 Comparing Data Lake vs Data Warehouse

For the PA company's big data project, a data lake is the preferred choice over a traditional data warehouse. Laskowski (2016) defines a data lake as a massively scalable storage repository that holds vast amounts of raw data in its native format until it is needed, along with processing systems that can ingest data without compromising the data structure. The primary justification lies in the heterogeneous nature of the data sources i.e, sensors, satellites, drones, social media, and market data which generate structured, semi-structured, and unstructured data formats, and data lakes are suitable to store such diverse data types as mentioned by Miloslavskaya and Tolstoy (2016) in their research on big data storage paradigms.

Terrizzano et al. (2015) emphasize that data lakes excel at storing raw data in its native format without requiring predefined schemas, offering the flexibility essential for agricultural big data applications. Given the 600+ petabyte volume, Fang (2015) notes that data lakes provide cost-effective storage using distributed file systems like Hadoop HDFS, which scales horizontally to accommodate exponential data growth. Furthermore, Khine and Wang (2018) explain that data lakes support schema-on-read approaches, enabling data scientists to define structures during analysis rather than during ingestion, which is crucial for exploratory analytics in precision agriculture.

Inmon (2016) contrasts this by highlighting that traditional data warehouses impose rigid schema-on-write constraints that hinder agility when integrating new sensor types or data sources, making data lakes the optimal foundation for the PA company's advanced agricultural analytics.

Task A.2: Graph Database Recommendation

For the PA company's requirement to query complex relationships between plants, crops, diseases, symptoms, and pests, a graph database represents the optimal analytical data store. Robinson et al. (2015) explain that graph databases such as Neo4j or Amazon Neptune are specifically designed to model and traverse interconnected entities through nodes and edges, making them ideal for relationship-heavy queries like "find all cotton diseases directly or indirectly caused by nitrogen deficiency".

Angles and Gutierrez (2008) highlight a critical limitation of traditional relational databases: they struggle with multi-level relationship queries, requiring expensive JOIN operations that degrade performance exponentially with relationship depth. In contrast, Robinson et al. (2015) describe how graph databases utilize index-free adjacency, where each node maintains direct references to adjacent nodes, enabling constant-time traversals regardless of dataset size a crucial advantage for agricultural knowledge graphs where disease causation may span multiple intermediate factors.

Francis et al. (2018) demonstrate that graph query languages like Cypher provide intuitive syntax for expressing pattern-matching queries across relationship paths, significantly simplifying the development of agricultural decision support systems. Furthermore, Angles (2012) explains that graph databases support property graphs where nodes and edges can store rich metadata about crop characteristics, environmental conditions, and disease symptoms, enabling sophisticated multi-attribute analysis. This semantic richness combined with efficient traversal capabilities makes graph databases the superior choice for the PA company's interconnected agricultural knowledge base.

Task A.3: MapReduce for Real-Time Analytics

I disagree with using MapReduce for the PA company's real-time prediction and analytics requirements. Dean and Ghemawat (2008) describe MapReduce as fundamentally a batch processing framework designed for high-throughput offline processing of large datasets, with typical job execution times measured in minutes to hours rather than seconds. Zaharia et al. (2010) highlight that the framework's architecture involves substantial overhead from job initialization, task scheduling, and disk-based intermediate data storage, making it unsuitable for latency-sensitive applications requiring responses within seconds.

For real-time agricultural analytics, Karimov et al. (2018) recommend stream processing frameworks such as Apache Kafka Streams, Apache Flink, or Apache Storm as appropriate alternatives, noting that these systems process data continuously as it arrives from sensors and satellites. Carbone et al. (2015) explain that stream processing systems maintain stateful computations in memory, enabling sub-second response times that MapReduce cannot achieve. Furthermore, Carbone et al. (2017) demonstrate that Flink provides exactly-once processing semantics with event-time processing capabilities essential for temporal agricultural data analysis.

Toshniwal et al. (2014) emphasize that modern precision agriculture applications require immediate responses to critical events such as frost warnings, irrigation alerts, or pest outbreak detection requirements that stream processing architectures satisfy through micro-batch processing or event-driven computation models. Therefore, whilst MapReduce remains valuable for historical analysis and model training, stream processing frameworks are indispensable for the PA company's real-time operational requirements.

2. Task B: Information Visualisation

Figure 1: Guest Composition

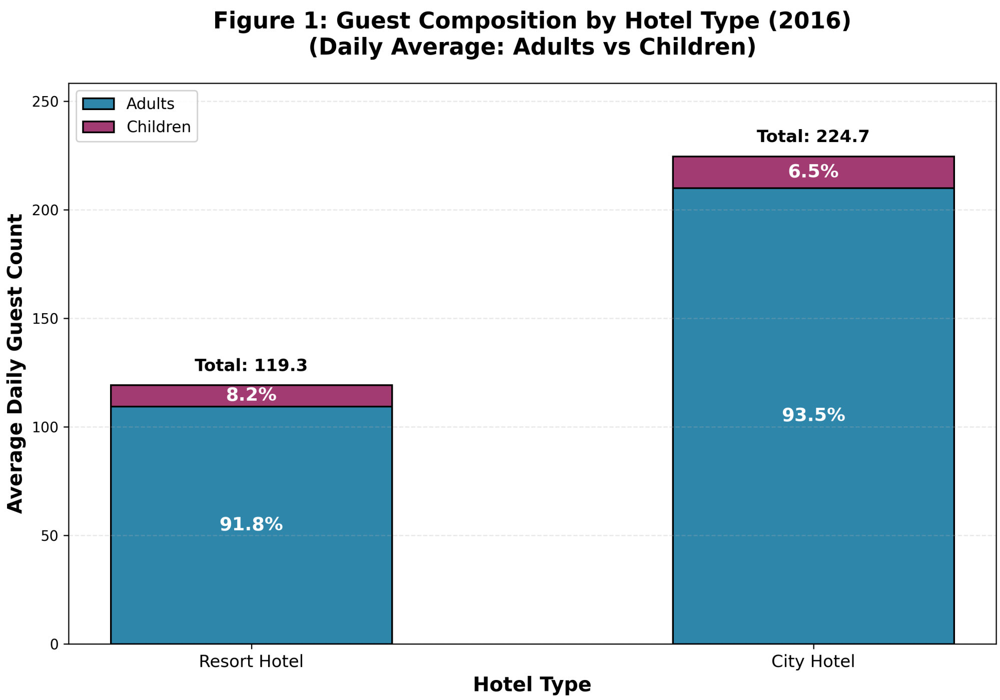

Figure 1: Guest Composition

Justification:

Figure 1 employs a stacked bar chart to compare guest composition between Resort Hotel and City Hotel, with bar heights representing actual daily guest volumes and internal segments showing percentage distributions of adults versus children. Stacked bar charts are particularly effective for simultaneously displaying both absolute magnitude and relative proportions within categorical data. The bars are scaled to actual average daily guest counts (119.3 for Resort Hotel and 224.7 for City Hotel), allowing immediate visual recognition of the substantial volume difference between properties. Blue segments represent adults while purple segments represent children, with percentage labels (91.8% and 8.2% for Resort Hotel, 93.5% and 6.5% for City Hotel) clearly indicating compositional differences. Total guest counts are displayed above each bar to reinforce the scale disparity. This visualization technique moves beyond simple proportion comparison by encoding two dimensions of information: the vertical position and height communicate absolute guest volumes, while the segmentation and percentage labels communicate demographic composition, providing stakeholders with comprehensive understanding of both operational scale and target market characteristics.

Key Insights:

The visualization reveals that City Hotel operates at substantially higher volume, averaging

224.7 total daily guests compared to Resort Hotel's 119.3, representing an 88.4% higher guest count. However, despite this volume disparity, both properties demonstrate remarkably similar adult-focused positioning, with City Hotel comprising 93.5% adults (210.0 guests) and only 6.5% children (14.7 guests), while Resort Hotel shows 91.8% adults (109.4 guests) and 8.2% children (9.8 guests). The proportional difference of 1.7 percentage points indicates Resort Hotel attracts marginally more families relative to its scale, though both properties remain predominantly adult-oriented. This composition pattern suggests both hotels prioritize business travelers and adult leisure segments, with children representing less than 10% of guest populations across both property types. The scale difference has significant operational implications: City Hotel requires proportionally larger infrastructure, staffing capacity, and service delivery systems to manage nearly double the daily guest volume, while Resort Hotel operates a more compact operation serving similar guest demographics at lower absolute numbers. The minimal compositional difference (1.7%) despite substantial volume difference (88.4%) indicates that family appeal is not the primary differentiator between these property types, instead suggesting that location, amenities, and service models drive the volume disparity while both properties serve fundamentally similar adult-dominated market segments.

Figure 2: Seasonal Booking Trends with 14-Day Rolling Average

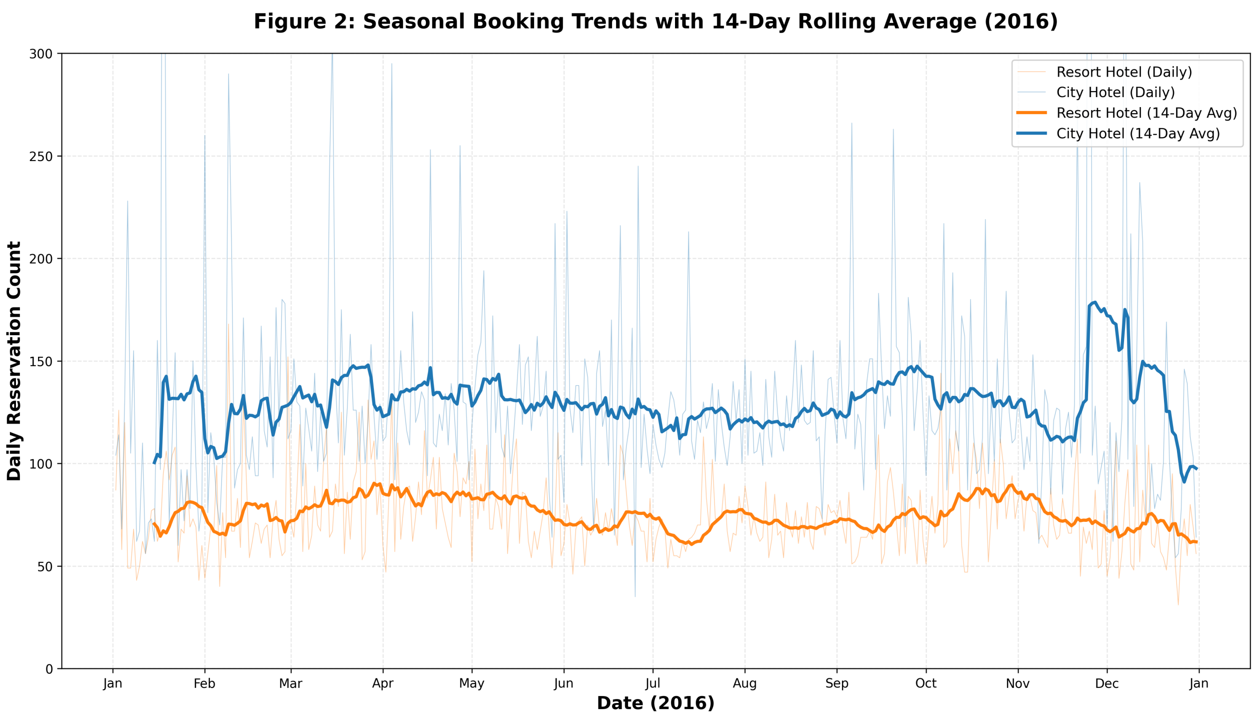

Figure 2: Seasonal Booking Trends with 14-Day Rolling Average

Justification:

Figure 2 utilizes a dual-layer line plot combining raw daily reservation data with 14-day rolling averages to reveal temporal booking patterns throughout 2016. Line plots are the standard technique for time-series data visualization, as they effectively communicate trends, seasonality, and cyclical patterns. The thin, semi-transparent lines (orange for Resort Hotel, blue for City Hotel) representing raw daily values preserve data granularity and expose day-to-day volatility, while thick, fully opaque lines in the same colors displaying 14-day rolling averages smooth short-term fluctuations to reveal underlying trends. This dual-layer approach balances detail preservation with interpretability, allowing readers to observe both individual daily performance and broader seasonal movements simultaneously. The two-week rolling average window is deliberately chosen to capture weekly cyclical patterns while filtering daily noise, providing meaningful trend visualization without excessive lag. Chronological ordering with monthly date markers on the x-axis enables intuitive navigation through the year, allowing stakeholders to identify peak periods, seasonal troughs, and comparative performance between hotel types across identical timeframes. The y-axis is capped at 300 reservations to optimize visual clarity for the majority of operational days.

Key Insights:

The visualization exposes contrasting temporal patterns between hotel types. Resort Hotel demonstrates relatively stable booking volumes throughout the year, averaging 75.0 daily bookings and peaking on February 9, 2016 with 168 reservations. Interestingly, the data reveals a slight negative 2.0% summer variance compared to winter months, indicating this particular resort property does not follow the typical leisure-driven seasonal pattern of summer surges. This atypical pattern may reflect the resort's specific geographic location, amenities portfolio, or target market segmentation that attracts consistent year-round demand rather than seasonal holiday travelers. Conversely, City Hotel maintains higher overall booking volumes with an annual average of 129.1 daily reservations and experiences a dramatic peak of 769 bookings on November 25, 2016. This exceptional spike likely corresponds to a major event, holiday shopping period, or conference activity, demonstrating the urban property's susceptibility to event-driven demand surges. The substantial difference in peak magnitudes (168 for Resort versus 769 for City) underscores fundamentally different operational challenges. City Hotel must maintain surge capacity for occasional extreme-volume days while Resort Hotel benefits from more predictable demand enabling consistent staffing and resource allocation strategies.

Figure 3: Hotel Operational Profile Comparison (Radar Plot)

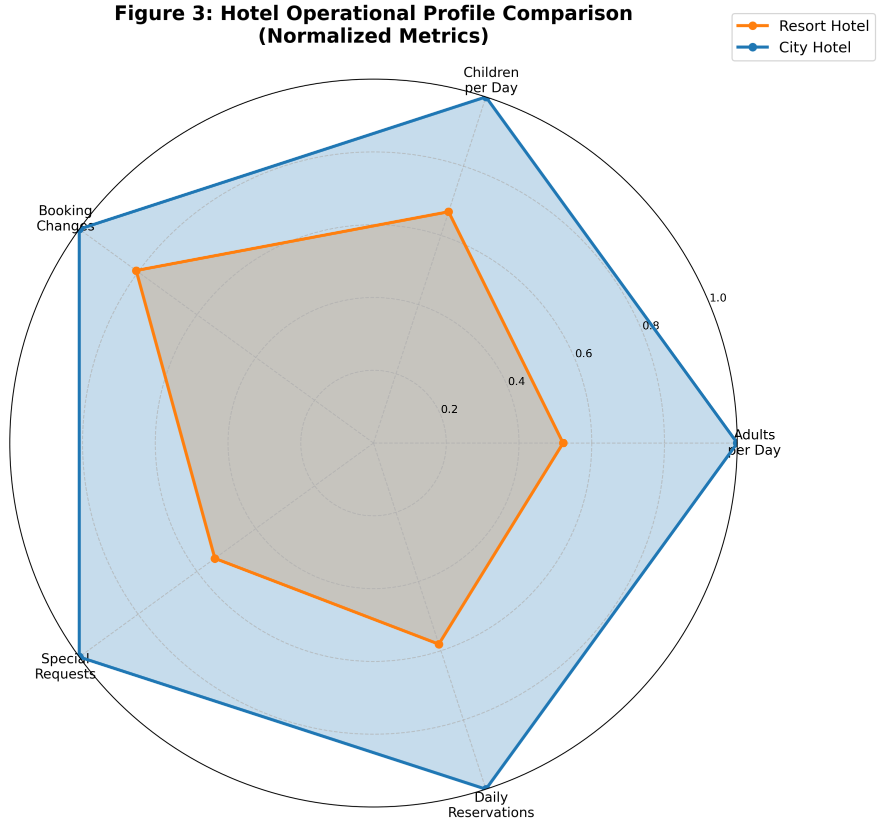

Figure 3: Hotel Operational Profile Comparison

Justification:

Figure 3 employs a radar plot to simultaneously compare five operational metrics across both hotel types on a normalized scale. Radar plots are specifically designed for multi-dimensional comparison, allowing viewers to assess multiple variables simultaneously and recognize overall operational profiles at a glance. Each axis radiating from the center represents a different metric: average adults per day, average children per day, average booking changes, average special requests, and average daily reservations. All values are normalized to a 0 to 1 scale using maximum values as benchmarks, ensuring fair comparison despite different measurement units and ranges. This standardization prevents metrics with naturally larger values from visually dominating the chart. The resulting shape formed by connecting plotted points with lines provides an immediate visual impression of each hotel's operational characteristics. Orange shading represents Resort Hotel while blue represents City Hotel, with semi-transparent fills enabling overlap visibility when profiles intersect. The circular grid lines marked at 0.2 intervals provide reference points for reading normalized values. This visualization technique synthesizes information from previous figures while adding operational complexity dimensions (booking changes and special requests), providing a holistic comparative view impossible to achieve through separate bar charts or tables without overwhelming cognitive load.

Key Insights:

The radar plot exposes fundamental operational differences between hotel types that extend beyond simple volume metrics. City Hotel demonstrates dominance across most dimensions, with actual values of 209.99 average adults per day, 129.08 average daily reservations, and 70.32 average special requests per day, achieving near-maximum normalized scores (0.9 to 1.0) for these metrics. This confirms its position as the higher-volume, adult-focused property established in earlier figures. Resort Hotel shows actual values of 109.45 average adults per day, 75.00 average daily reservations, and 37.91 average special requests per day. Interestingly, children counts are more comparable between hotels (Resort Hotel: 9.82, City Hotel: 14.69) than volume metrics, with Resort Hotel achieving 67% of City Hotel's children count while only achieving 58% of its adult count, reinforcing Resort Hotel's proportionally stronger family appeal. Booking changes show similar proportional patterns, with Resort Hotel averaging 21.27 changes per day versus City Hotel's 26.37, indicating complexity scales somewhat with volume but not perfectly. The contrasting shapes reveal that City Hotel's profile stretches prominently toward all axes due to higher absolute volumes, while Resort Hotel's more compact profile indicates lower operational scale but similar complexity-to-volume ratios. This suggests both properties manage comparable operational challenges relative to their respective scales, though City Hotel handles substantially higher absolute workloads requiring proportionally greater staffing and resource allocation.

Figure 4: Correlation Matrix of Booking Variables

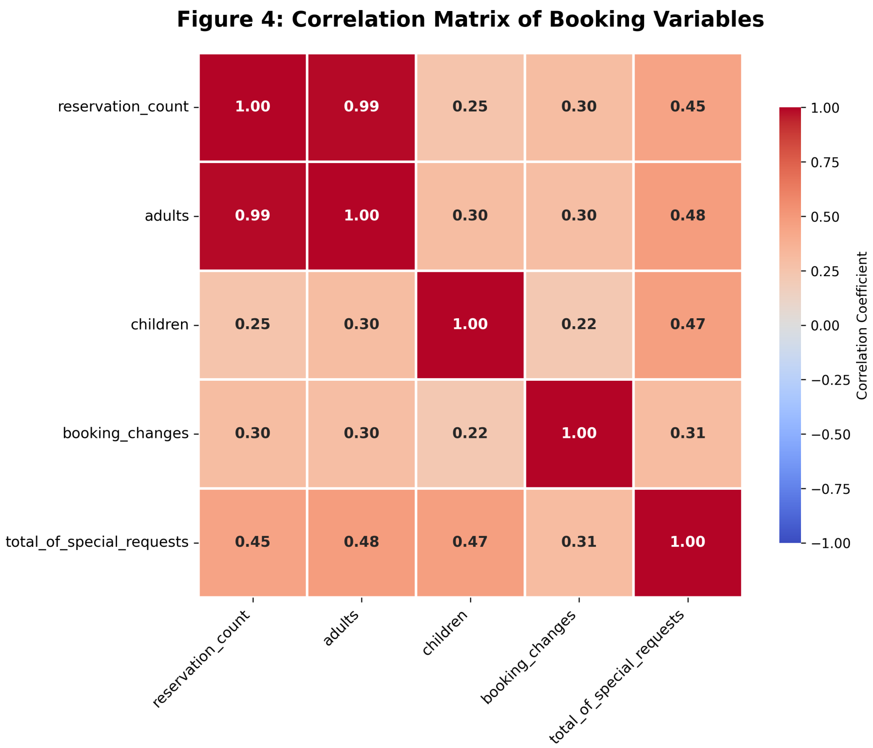

Figure 4: Correlation Matrix

Justification:

Figure 4 employs a correlation heatmap to systematically examine relationships among five key booking variables: reservation count, adults, children, booking changes, and special requests. Correlation heatmaps utilize color encoding to represent the strength and direction of linear relationships between variable pairs, with correlation coefficients ranging from negative 1 (perfect negative correlation, shown in blue) through 0 (no correlation, shown in white) to positive 1 (perfect positive correlation, shown in red). Each cell displays the Pearson correlation coefficient calculated across all operational days in the dataset, providing quantitative measures of relationship strength. This matrix format enables simultaneous examination of all possible variable pairings, revealing patterns that might not be apparent through individual bivariate analyses. The symmetrical arrangement along the diagonal (where each variable correlates perfectly with itself at 1.00) provides visual reference points, while off-diagonal cells expose meaningful operational relationships. This visualization technique is particularly valuable for exploratory data analysis, as it systematically identifies which variables move together, enabling data-driven decisions about which relationships warrant deeper investigation through subsequent visualizations such as scatter plots. The heatmap serves as an analytical roadmap, guiding stakeholders toward understanding the underlying structure of booking behavior patterns.

Key Insights:

The correlation matrix reveals several significant relationships with important operational implications. The strongest correlation exists between reservation count and adults (r equals 0.987), indicating an almost perfect linear relationship where daily booking volume directly predicts total adult guest count. This exceptionally strong correlation suggests these metrics are nearly interchangeable for capacity planning purposes, as each additional reservation contributes consistently to adult guest volumes. Adults demonstrate moderate positive correlations with both special requests (r equals 0.478) and children (r equals 0.298), suggesting larger adult counts are associated with increased service demands and modest increases in children, though neither relationship is deterministic. Special requests show moderate correlations with multiple variables including adults (0.478), children (0.473), and reservation count (0.448), indicating service demand scales somewhat predictably with guest volumes but maintains substantial independence, as correlation coefficients below 0.5 suggest other factors drive approximately 75% of special request variation. Notably, booking changes demonstrate the weakest correlations across all variables, with coefficients ranging from 0.220 to 0.308, confirming that modification behavior operates largely independently from guest composition and service demands. This independence pattern has critical planning implications: hotels cannot reliably predict booking modification workload from reservation volumes or guest demographics, necessitating separate capacity planning for modification processing systems distinct from front-desk service delivery operations. The matrix structure reveals that while volume metrics (reservations, adults) cluster together with strong intercorrelation, complexity metrics (changes, requests) maintain relative independence, suggesting hotels face two distinct operational challenge domains requiring differentiated management approaches rather than unified scaling strategies.

Figure 5: Relationship Between Party Size and Service Demands

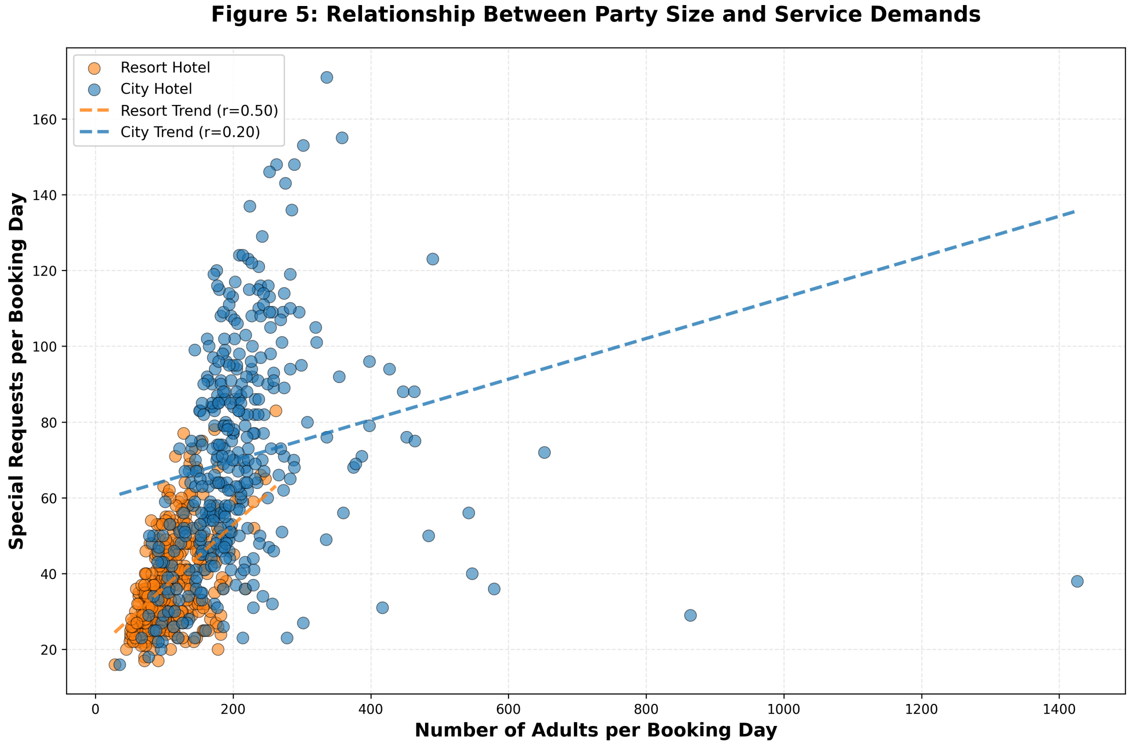

Figure 5: Relationship Between Party Size and Service Demands

Justification:

Figure 5 employs a scatter plot to investigate the relationship between daily adult guest counts and special service requests, utilizing bivariate visualization to expose whether larger guest volumes generate proportionally increased service demands. Scatter plots are the standard technique for examining correlations between continuous variables, as the spatial distribution of points immediately reveals relationship patterns, strength, and outliers. Each point represents a single operational day, with horizontal position indicating total adult count that day and vertical position showing special request volume. Orange circles denote Resort Hotel days while blue circles represent City Hotel days, enabling simultaneous examination of whether this relationship differs by property type. Dashed trend lines generated through linear regression provide visual summary of overall patterns, with slopes indicating how service requests change per additional adult guest. Correlation coefficients (r-values) displayed in the legend quantify relationship strength, with Resort Hotel showing r equals 0.464 and City Hotel r equals 0.458, indicating moderate positive correlations. This analytical approach transcends simple averages by exposing the full distribution of outcomes, revealing that while general positive trends exist, substantial variability persists with some high-adult days generating modest service requests and some lower-adult days producing elevated demands, suggesting additional factors beyond volume influence service complexity.

Key Insights:

The visualization reveals moderate positive correlations between adult counts and special requests for both properties (Resort Hotel r equals 0.464, City Hotel r equals 0.458), with trend line equations showing Resort Hotel generates approximately 0.31 additional requests per additional adult (y equals 0.306x plus 4.121) while City Hotel generates 0.28 additional requests per adult (y equals 0.277x plus 11.862). These similar correlation strengths and slopes indicate both properties experience comparable service demand scaling patterns regardless of operational volume differences, suggesting service intensity per guest operates consistently across property types. The moderate correlation magnitudes (approximately 0.46) indicate adult count explains roughly 21% of service request variation (r-squared), meaning 79% of service demand variation stems from factors other than simple guest volume, such as guest demographics, booking channels, length of stay, or seasonal travel purposes. The scatter pattern shows most operational days cluster in predictable bands along trend lines, with Resort Hotel days typically ranging from 80 to 140 adults generating 30 to 60 requests, while City Hotel days spanning 150 to 250 adults produce 50 to 100 requests. However, notable outliers exist: some Resort Hotel days with only 100 adults generate 70 plus requests (high-maintenance guests), while some City Hotel days with 200 adults produce fewer than 60 requests (low-touch travelers). These outliers have critical staffing implications, as hotels cannot rely solely on booking forecasts to predict service workload. Management must maintain surge capacity for high-service-intensity days that exceed predictable patterns, suggesting flexible staffing models and cross-training initiatives are essential for maintaining service quality during unexpectedly demanding periods despite moderate overall correlation between volume and service complexity.

Critical Review

This visualization project demonstrated practical application of information visualization principles, particularly the fundamental tenet that visualization technique selection must align with data type and analytical objectives. The progression from proportion comparison (bar chart) through temporal analysis (line plots) to correlation investigation (scatter plot) and multi-dimensional profiling (radar plot) reflects deliberate structuring principles that guide readers through increasingly sophisticated analytical layers. Best practices were demonstrated through consistent color encoding (orange for Resort Hotel, blue for City Hotel throughout all figures), appropriate use of normalization in the radar plot to enable fair comparison across different measurement scales, and dual-layer temporal visualizations balancing granularity with interpretability through combining raw data with rolling averages. The decision to cap y-axes in Figures 2 and 3 at 300 exemplifies the practical tension between data completeness and visual effectiveness. While statistically comprehensive representations would include all extreme values (such as City Hotel's 769-booking peak day), such outliers compress the scale and obscure patterns affecting the majority of operations. One limitation identified is the scatter plot's density in Figure 5, where overlapping points potentially obscure distribution patterns. Alternative techniques such as hexbin plots or contour density visualization might improve clarity. The emphasis on audience-centered design proved valuable. Avoiding technical jargon, explaining visual encoding choices (solid versus dashed lines, normalized scales, correlation coefficients), and providing explicit descriptions ensures accessibility for stakeholders without data visualization expertise, fulfilling the communication purpose fundamental to effective business analytics.

Summary Conclusions

I. City Hotel operates at substantially higher scale, serving 224.7 average daily guests compared to Resort Hotel's 119.3 guests (88.4% difference), though both properties demonstrate similar adult-focused positioning with City Hotel comprising 93.5% adults and Resort Hotel 91.8% adults, indicating minimal compositional differences despite significant volume disparities.

II. Temporal analysis reveals contrasting volatility patterns: Resort Hotel maintains stable year-round demand averaging 75.0 daily bookings with minimal seasonal variation (negative 2.0% summer variance), while City Hotel averages 129.1 daily bookings with dramatic event-driven peaks reaching 769 reservations on November 25, necessitating fundamentally different capacity management strategies.

III. Correlation matrix analysis exposes strongest relationship between reservation count and adults (r equals 0.987), confirming volume metrics are nearly interchangeable for planning purposes, while special requests demonstrate moderate correlations with adults (0.478), children (0.473), and reservation count (0.448), indicating service demand scales somewhat predictably with guest volumes but maintains substantial independence.

IV. Scatter plot analysis reveals moderate positive correlations between adult counts and special requests for both properties (Resort Hotel r equals 0.464, City Hotel r equals 0.458), with trend lines indicating approximately 0.31 additional requests per additional adult for Resort Hotel and 0.28 for City Hotel, enabling predictable staffing ratios though 79% of service demand variation stems from factors beyond simple guest volume.

V. Radar plot visualization confirms City Hotel dominates across all operational dimensions (209.99 adults per day, 70.32 special requests, 26.37 booking changes, 14.69 children, 129.08 reservations) compared to Resort Hotel (109.45 adults, 37.91 requests, 21.27 changes, 9.82 children, 75.00 reservations), though proportional complexity-to-volume ratios remain comparable, suggesting both properties manage similar operational challenges relative to their respective scales.

VI. These findings demonstrate distinct operational models: City Hotel pursues volume-based efficiency managing nearly double the guest load with proportionally scaled service infrastructure, while Resort Hotel operates at lower absolute volumes with marginally higher family appeal (8.2% versus 6.5% children), requiring differentiated resource allocation strategies aligned to their respective volume profiles, volatility patterns, and guest demographic compositions.

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